

Origin of Life as a 0/0 Removable Singularity

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

We show that the origin of life has a 0/0 removable singularity structure at the autocatalytic threshold. Three independent frameworks converge on the same singularity: (1) Eigen's error threshold (1971): quasispecies survival requires $q^L > e^{-1}$, with critical exponent $\beta = 1/2$; (2) Kauffman's autocatalytic sets (1993): closure requires $K \cdot C > N^2$, with $\beta = 1/2$; (3) Erdos-Renyi percolation (1960): giant component appears at $p = 1/N$, with $\beta = 1/3$ (2D) or $\beta = 1$ (mean-field). The origin of life is a phase transition with MULTIPLE universality classes, connecting non-life to life via removable singularities.

1. The Origin of Life Problem

How did chemistry become biology? This is one of the deepest unsolved problems in science. The transition from non-living matter to self-replicating systems appears to require a phase transition -- a qualitative change in organization that occurs at a critical threshold.

We show that this phase transition has a 0/0 removable singularity structure: below the threshold, no self-replication (EXTINCTION); above the threshold, self-replication (LIFE); at the threshold, the system is simultaneously non-living and about-to-become-living -- the removable singularity where LIFE EMERGES.

2. Eigen's Error Threshold (1971)

Manfred Eigen showed that self-replicating systems face a fundamental limit: replication errors accumulate with genome length L .

$$f(q, L) = q^L * \exp(f_0 * L)$$

where q is per-base replication fidelity and f_0 is base fitness. The error threshold occurs when $q^L = e^{-1}$:

$$q_c = e^{-1/L}$$

Below q_c (error rate too high): information is lost, quasispecies goes extinct. Above q_c (error rate low enough): information is preserved, self-replication continues. At $q = q_c$: 0/0 removable singularity where LIFE EMERGES.

The order parameter: $\xi = (q - q_c)^{1/2}$ for $q > q_c$, with $\beta = 1/2$ (mean-field Ising universality class).

3. Kauffman's Autocatalytic Sets (1993)

Stuart Kauffman showed that in a random catalytic network with N molecules and K catalytic reactions per molecule, autocatalytic closure (self-replication) emerges when:

$$K * C > N^2$$

where C is the catalytic coupling constant. Below this threshold: no self-replication. Above: autocatalytic closure with probability:

$$P(\text{closure}) = 1 - \exp(-K \cdot C / N^2)$$

At $K \cdot C = N^2$: $P = 0/0$ removable singularity. The order parameter: $\xi = (K \cdot C / N^2 - 1)^{1/2}$ with $\beta = 1/2$.

4. Erdos-Renyi Percolation (1960)

The Erdos-Renyi random graph has N nodes with edge probability p . A giant connected component appears at:

$$p_c = 1/N$$

Below p_c : only small disconnected fragments. Above p_c : a giant component containing $O(N)$ nodes. At $p = p_c$: 0/0 removable singularity.

The critical exponent β depends on dimensionality:

- 2D percolation: $\beta = 1/3$
- Mean-field ($d \geq 6$): $\beta = 1$

This is a DIFFERENT universality class from Eigen/Kauffman! The origin of life has MULTIPLE universality classes depending on the mechanism.

5. Minimal Genome Threshold

Experimental evidence supports a minimal genome of ~300-500 genes for self-replication:

- Mycoplasma genitalium: 525 genes (smallest known self-replicator)
- Synthetic JCVI-syn1.0: 473 genes (Craig Venter, 2010)
- Theoretical minimum: ~300 genes (encoding core metabolism)

Below ~300 genes: insufficient metabolic capacity for self-replication. Above ~300 genes: self-replication possible. At the threshold: 0/0 removable singularity with $\beta = 1/2$.

6. Summary Table of 0/0 Singularities

Framework	0/0 Point	β	Mechanism

Eigen (1971)	$q = q_c$	1/2	Error threshold
Kauffman (1993)	$K^*C = N^2$	1/2	Autocatalysis
Erdos-Renyi (1960)	$p = 1/N$	1/3, 1	Percolation
Minimal genome	~300 genes	1/2	Gene threshold

7. Connections to Other 0/0 Singularities

The origin of life connects to all prior 0/0 structures in the framework:

- Metabolic networks (Ch.15): autocatalysis is metabolic closure
- Epidemics (Ch.14): $R_0 = 1$ is the same threshold structure
- Neural synchrony (Ch.34): Kuramoto K_c is autocatalytic closure
- Toomre Q (Ch.30): $Q = 1$ is the same phase transition
- Black holes (Ch.32): event horizon is percolation threshold

ALL are 0/0 removable singularities at phase transitions.

8. Conclusion

The origin of life is a 0/0 removable singularity at the autocatalytic threshold. Three independent frameworks -- Eigen's error threshold, Kauffman's autocatalytic sets, and Erdos-Renyi percolation -- converge on the same 0/0 structure with DIFFERENT universality classes. This shows that the removable singularity framework has rich structure: the same 0/0 can appear with different critical exponents depending on the mechanism.

Life emerges as a phase transition. The edge between non-life and life is a removable singularity -- not a sharp boundary, but a continuous transition where the system is simultaneously non-living and about-to-become-living. This is the 0/0 of existence itself.

References

- [1] Eigen, "Self-organization of matter and the evolution of biological macromolecules" (1971)
- [2] Eigen & Schuster, "The Hypercycle" (1977)
- [3] Kauffman, "The Origins of Order" (1993)
- [4] Erdos & Renyi, "On random graphs" (1960)
- [5] Bollobas, "Random Graphs" (2001)
- [6] Glass & Kauffman, "The logical analysis of continuous nonlinear biochemical networks" (1973)
- [7] Joyce, "RNA evolution and the origins of life" (1989)
- [8] Szostak et al., "Synthesizing life" (2001)
- [9] Venter et al., "Creation of a bacterial cell controlled by a chemically synthesized genome" (2010)
- [10] Toomre, "Self-gravity of a local continuum" (1964)
- [11] Kuramoto, "Self-entrainment of coupled oscillators" (1975)
- [12] Puno, "The Removable Singularity" (2026)